

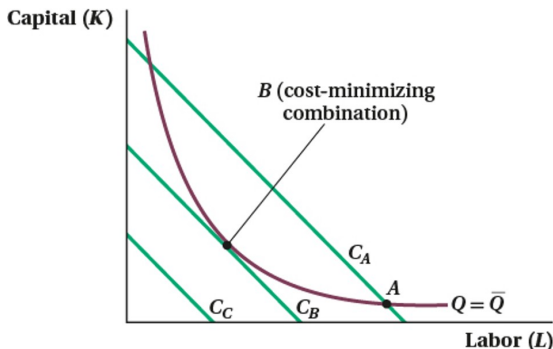
Microeconomic Theory: TA Session

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Cost minimization



$$MRTS = \frac{MP_L}{MP_K} = \frac{w}{r}$$

Interpretation: the ratio of marginal product should be equal to the ratio of input “prices”

$$\text{Rearranging, we have } \frac{MP_L}{w} = \frac{MP_K}{r}$$

Interpretation: Each input should have the same marginal productivity relative to its price

Cost minimization - exercise

Suppose a firm's production function is $Q = K^{0.5}L^{0.5}$, the rental rate of capital is $r = 1$, and the wage for labor is $w = 4$.

1. Find the cost minimizing K and L to produce a quantity of 20.

$$\text{Answer: } \frac{MP_L}{MP_K} = \frac{w}{r} \Rightarrow \frac{0.5K^{0.5}L^{-0.5}}{0.5K^{-0.5}L^{0.5}} = \frac{K}{L} = \frac{4}{1}, \text{ so } K = 4L.$$

$$\begin{aligned} \text{To produce 20 units: } Q &= K^{0.5}L^{0.5} = (4L)^{0.5}L^{0.5} = 2L = 20 \\ \Rightarrow L &= 10, K = 40 \end{aligned}$$

2. What is the minimized total cost?

$$\text{Answer: Total cost} = wL + rK = 4 \times 10 + 1 \times 40 = 80.$$

Homework 4

4.3. The factory's production function: $q = L^{2/3}K^{1/3}$. Input prices: $w = r = 25$. Units can also be produced by Artisans according to $g(A) = 20A - 2A^2$, and the unit price of A is 1.

- (c) What is the minimum cost choice of (L, K, A) that produces 144 units?

Answer: First, notice that the optimal choice will have $\frac{MP_L}{w} = \frac{MP_K}{r} = \frac{MP_A}{p_A}$.

The cost minimizing condition of the factory: $\frac{MP_L}{w} = \frac{MP_K}{r}$
 $\Rightarrow \frac{\frac{2}{3}L^{-1/3}K^{1/3}}{25} = \frac{\frac{1}{3}L^{2/3}K^{-2/3}}{25} \Rightarrow L = 2K.$

Plugging this into $\frac{MP_L}{w}$, we have $\frac{MP_L}{w} = \frac{\frac{2}{3}(2K)^{-1/3}K^{1/3}}{25} = \frac{2^{2/3}}{75}.$

Then we use $\frac{MP_L}{w} = \frac{MP_A}{p_A}$ to get: $\frac{2^{2/3}}{75} = \frac{20-4A}{1}$. Solving, we have

$$A = \frac{20 - \frac{2^{2/3}}{75}}{4} \approx 4.995.$$

Artisan production is $20A - 2A^2 \approx 50$, so the factory produces $144 - 50 = 94$, and $q = L^{2/3}K^{1/3} = 2^{2/3}K = 94 \Rightarrow K \approx 59.2$, $L \approx 118.4$.

Cost curves

A firm's total cost function is $TC(Q) = 100 + 10Q + Q^2$.

- Derive MC, FC, AVC, and ATC.

Answer:

$$MC = 10 + 2Q, \quad FC = 100, \quad VC = 10Q + Q^2, \quad AVC = \frac{VC}{Q} = 10 + Q, \quad ATC = \frac{TC}{Q} = \frac{100}{Q} + 10 + Q.$$

- Find the output level where ATC is minimized.

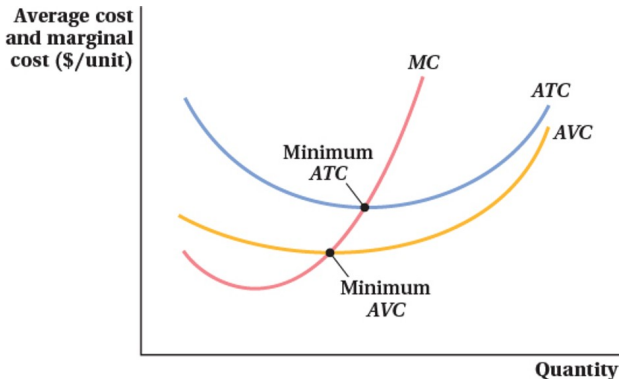
Answer: ATC is minimized where $MC = ATC$:

$$10 + 2Q = \frac{100}{Q} + 10 + Q \Rightarrow Q = 10.$$

Cost curves

Why does MC cross ATC where ATC is the lowest?

Answer: When MC is below ATC, each additional (“marginal”) unit of production will drag ATC down; when MC is above ATC, each marginal unit of production will drag ATC up. So, MC crosses ATC only when the slope of ATC is flat — that is to say, when ATC is lowest.

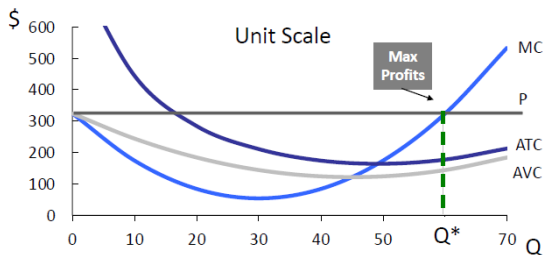


Profit maximization

A firm's profit is maximized at the quantity where $MC = MR$

- ▶ Before that, $MC < MR$, so there is still profit to be made from producing additional units
- ▶ But from the $MC = MR$ point onwards, every additional unit of production loses money because the firm is going into $MC > MR$ territory

The graph below is for a price-taking firm (where $P=MR$)



The decision to shut down

In the short run, the firm still incurs fixed costs even if it shuts down

A firm chooses to shut down in the short run if:

$$\begin{aligned}TR &< VC \\ \implies TR/Q &< VC/Q \\ \implies P &< AVC\end{aligned}$$

In the long run, the firm no longer incurs fixed cost if it shuts down

A firm chooses to shut down in the long run if:

$$\begin{aligned}TR &< TC \\ \implies TR/Q &< TC/Q \\ \implies P &< ATC\end{aligned}$$

The decision to shut down - exercise

A firm in a perfectly competitive market has total cost function $TC = 100 + 10q + 2q^2$.

1. With a price of 30, determine whether the firm should produce or shut down in the short run.

Answer: Profit maximization: $30 = P = MC = 10 + 4q \Rightarrow q = 5$.

In the short run, we compare P with AVC .

$VC = 10q + 2q^2$, so $AVC = 10 + 2q = 20 < P$, the firm will produce in the short run.

2. If the price and cost function does not change, determine whether the firm should produce or shut down in the long run.

Answer: In the long run, we compare P with ATC .

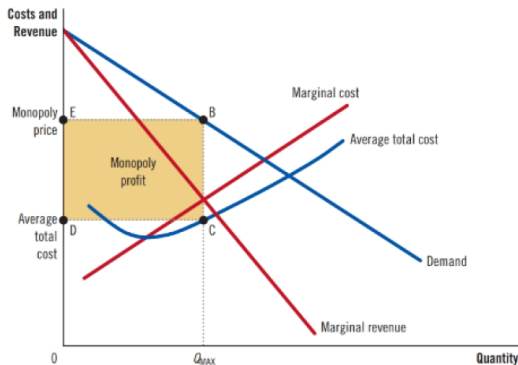
$ATC = 100/q + 10 + 2q = 40 > P$, the firm will shut down in the long run.

Monopoly

In a monopoly, the firm's demand curve is the market demand curve, so $P > MR$.

The monopolist's profit is maximized when $MR=MC$. It can earn a positive profit.

- ▶ The monopolist's equilibrium quantity is smaller than under perfect competition ($P=MC$), so there is deadweight loss



Monopoly - exercise

Suppose Hopkins Cafe runs a monopoly of grilled chicken thighs in the on-campus dining market. The market demand curve for grilled chicken thighs is $P = 24 - Q$, and the total cost curve for Hopkins Cafe is $TC = Q^2 + 64$.

1. Derive Hopkins Cafe's MR curve.

Answer: $P = 24 - Q$, revenue = $PQ = 24Q - Q^2$, $MR = 24 - 2Q$.

2. Why does the MR curve has the same intercept as the demand curve? Provide intuition.

Answer: Because, at the intercept, $Q = 0$. When the monopolist increase production, it doesn't need to lower the price for existing production (because there isn't any existing production).

3. Calculate the equilibrium quantity and price, and the profit of the firm.

Answer: $MC = TC' = 2Q$, $MR = MC \Rightarrow 24 - 2Q = 2Q$
 $\Rightarrow Q = 6$, $P = 24 - 6 = 18$. The profit of the firm is
 $PQ - TC = 18 \times 6 - (6^2 + 64) = 8$.

Monopoly - exercise

Suppose Hopkins Cafe runs a monopoly of grilled chicken thighs in the on-campus dining market. The market demand curve for grilled chicken thighs is $P = 24 - Q$, and the total cost curve for Hopkins Cafe is $TC = Q^2 + 64$.

4. If, instead, this is a perfectly competitive market (with the same cost and market demand curves), what is the price and quantity sold?

Answer: In a perfectly competitive market, equilibrium is given by $P = MR = 24 - Q$, $MR = MC \Rightarrow 24 - Q = 2Q \Rightarrow Q = 8, P = 16$

Monopoly - exercise

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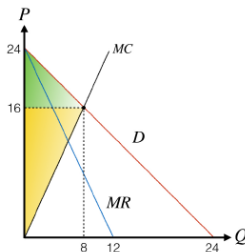
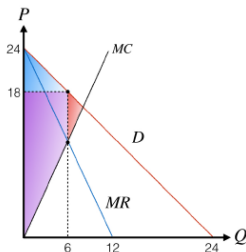
5. Calculate the consumer and producer surplus under monopoly and perfect competition.

Answer:

Under monopoly, CS is the blue area: $CS = (24 - 18) \times 6/2 = 18$. PS is the purple area: $PS = (18 + 6) \times 6/2 = 72$.

Under perfect competition, CS is the green area:

$CS = (24 - 16) \times 8/2 = 32$. PS is the yellow area: $PS = 16 \times 8/2 = 64$



Monopoly - exercise

Suppose Hopkins Cafe runs a monopoly of grilled chicken thighs in the on-campus dining market. The market demand curve for grilled chicken thighs is $P = 24 - Q$, and the total cost curve for Hopkins Cafe is $TC = Q^2 + 64$.

6. From the previous question, we see that monopoly is inefficient. What is the deadweight loss from monopoly?

Answer: Total surplus under perfect competition is $32 + 64 = 96$. Total surplus under monopoly is $18 + 72 = 90$. The deadweight loss is their difference: $96 - 90 = 6$.

Alternatively, we can directly calculate the deadweight loss from the red area.

